

12 INSTALLATION

12.1 Summary

This chapter describes the installation of the device.

The installation and test operation of the device are performed by Ishida service engineers. Before the installation, check the installation location and prepare the power source.

Additionally follow instructions in this chapter when relocating or moving the device.

WARNING

- **The installation and test operation of the device can be performed only by the Ishida service department.**

CAUTION

- **When relocating the device, contact the distributor or Ishida customer support.
When the installation is not appropriate, accurate weighing may not be achieved.**

<Contents>

- Installation conditions, transportation and lifting, and installation procedures

<Purpose>

- To understand the appropriate installation environment, safe transport and relocation of the device, and secure installation procedures.

<Intended reader>

- Maintenance engineers
- Ishida service engineers

12.2 Installation Condition

When installing the device, satisfy the following installation conditions in order to operate the device effectively, maintain precision and perform weighing safely.

WARNING

- The installation and test operation of the device can be performed only by the Ishida service department.

CAUTION

- When relocating the device, contact the distributor or Ishida Customer Support.
- When installing the device, observe the installation conditions. Inappropriate installation may cause malfunction or damage to the device.

NOTE

- The power source specifications vary depending on customer. Observe the power specifications in the contract.
- The device weight varies depending on the customer. Refer to the specifications issued at the order.

Table 12-1

Item	Condition
Operation environment	Avoid direct sunlight. • Room temperature: 0 to 40°C. • Room humidity: 35 to 85% R.H (No condensation is allowed.) No dust is allowed. There should be sufficient light to take product photos with a camera.
Installation surface	The floor should have a steady, horizontal surface without vibrations.
Installation space	There should be a working space around the device for maintenance and inspections.
Static electricity	Make sure to install the parts appropriately in order to avoid unearthed metal pieces.
Radio interference	Do not place the device close to radio equipment or sources of radio interference.
Air (For WP specifications)	Dry air. Pressure: 0.5 MPa to 0.6 MPa. (5 kgf/cm ² to 6 kgf/cm ²)
Power (For standard specifications)	200 to 240 V, 50/60 Hz (Single phase) Voltage fluctuation: within $\pm 10\%$. Frequency fluctuation: within $\pm 5\%$. Do not share the power source with devices that may emit noises.

Table 12-1

Item	Condition
Weight (For standard specifications)	500 kg.

12.3 Transportation and Lifting

This section describes the precautions for the transporting and lifting of the weighing machine and lifting procedures. Understand them beforehand in order to avoid accidents during the installation of the weighing machine.

WARNING

- The installation and test operation of the device can be performed only by the Ishida service department.
- When transporting or lifting the weighing machine, always observe the following safety precautions, customer safety standard, and relevant national and local laws.
If they are not observed, injuries, fatal accidents or damage to the weighing machine may occur.

CAUTION

- When relocating the device, contact the distributor or Ishida Customer Support.

12.3.1 Precaution for Transporting

When transporting the weighing machine, observe the following items:

1. Place a "match mark" on the pipes, connections, and fitting portions, and do not remove the "match mark" during transportation.
2. Seal the pipes and openings with removable sealing caps, and remove the caps on site immediately before reassembling. Protect the protruding part of the male screws with cloth or similar material.
3. When it is necessary to disconnect the electric cables for transportation, tie the cable ends by groups, cover with vinyl or similar material for protection, and transport to the site. Remove the cover immediately before the installation of the weighing machine.
4. When moving the packaged weighing machine by forklift, securely insert the fork into the specified positions.

12.3.2 Precaution for Lifting

When lifting the weighing machine, observe the following items.

NOTE

- For the weight of the weighing machine, refer to the specifications issued at the order. For the weight when the optional specifications are additionally installed, contact the distributor or Ishida customer support.

1. Take into account the weight of the weighing machine, and the center of gravity when lifting.
2. Select the lifting equipment within the safety load limit in accordance with the weight of the weighing machine.
3. Check items before lifting
 - There should be no twist or damage on the wire ropes.
 - The wire ropes should have a sufficient length.
 - The eyebolts should be securely installed.
4. Check items during lifting
 - Only a licensed personnel should perform the slinging work and operate a lifting weighing machine.
 - Select and use an appropriate lifting weighing machine in accordance with the load weight.
 - Apply patches or cloth for the protruding parts and edges in order to avoid damage to the weighing machine.
 - When starting the lifting work, give a sign to the other workers on the site.
 - Check the inclination and balance of the load while lifting the weighing machine.
 - Check the conditions of the lifting equipment and wire ropes while lifting the weighing machine.
5. Precautions for moving while lifting
 - Stay out from under the load.
 - Do not move the lifted load above the aisles.
6. Precautions for unloading
 - Remove objects from the unloading area, and observe the specified lifting point and slinging methods.

12.3.3 Weight of Device, Center of Gravity, and Dimensions

The center of gravity of the weighing machine is as follows.

NOTE

- For details such as the weight and dimensions of the weighing machine, refer to the specifications issued for the contract, or contact the distributor or Ishida customer support. When lifting the weighing machine, use an appropriate lifting weighing machine in accordance with the weighing machine weight, and take into account the center of gravity of the load when lifting.

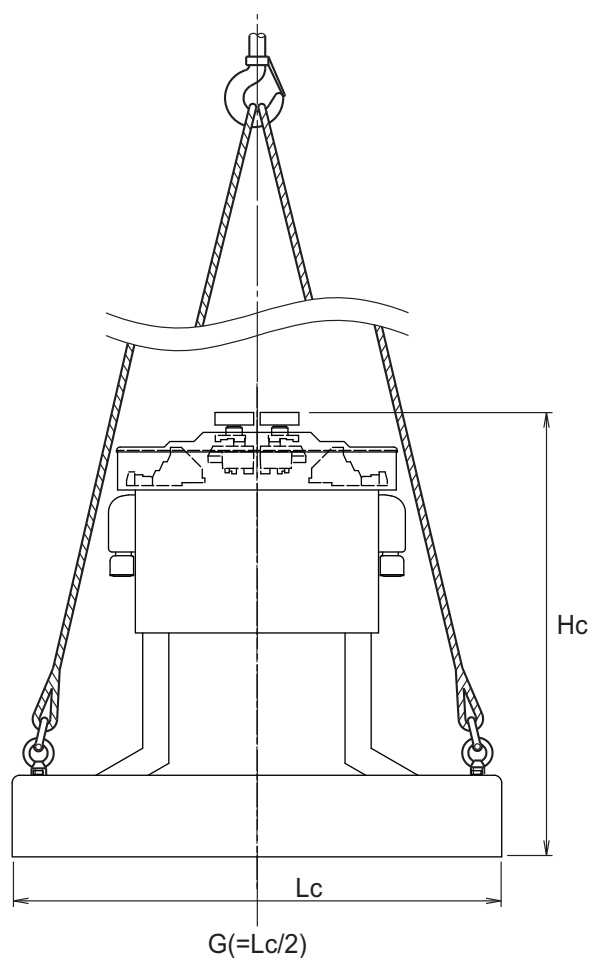


Fig. 12-1

12.3.4 Lifting Procedure

Lift the weighing machine in accordance with the following procedures.

WARNING

- **Make sure that there are no other employees in the vicinity of the weighing machine that is to be lifted.**
- **Stay out from under the lifted weighing machine.**

1. Make sure the hook is positioned above the center of gravity of the unit.
2. Before lifting the load, instruct all employees in the vicinity to stay away from the lifting site.
3. Inform the co-operator of the lift initiation with an arm signal.

CAUTION

- **Slowly lift the weighing machine. If the load is lifted quickly, it may damage the weighing machine and the lifting equipment such as the wire ropes.**

4. Begin the lift by removing slack from the lift cables.
5. Continue the lift until the weighing machine is off the ground. Check for inclination of the load. If the load is inclined, lower the weighing machine to the ground, eliminate the cause of the inclination, adjust and lift again.

NOTE

Possible causes of load inclination are below:

- Uneven wire length
- Inappropriate positioning between the center of gravity and wire rope lifting point

6. During the lifting of the weighing machine, check for failures of the wire ropes and lifting equipment, and load inclination.
7. After the unloading is complete and the eyebolts are removed, install the bolt hole caps.

12.4 Installation Procedure

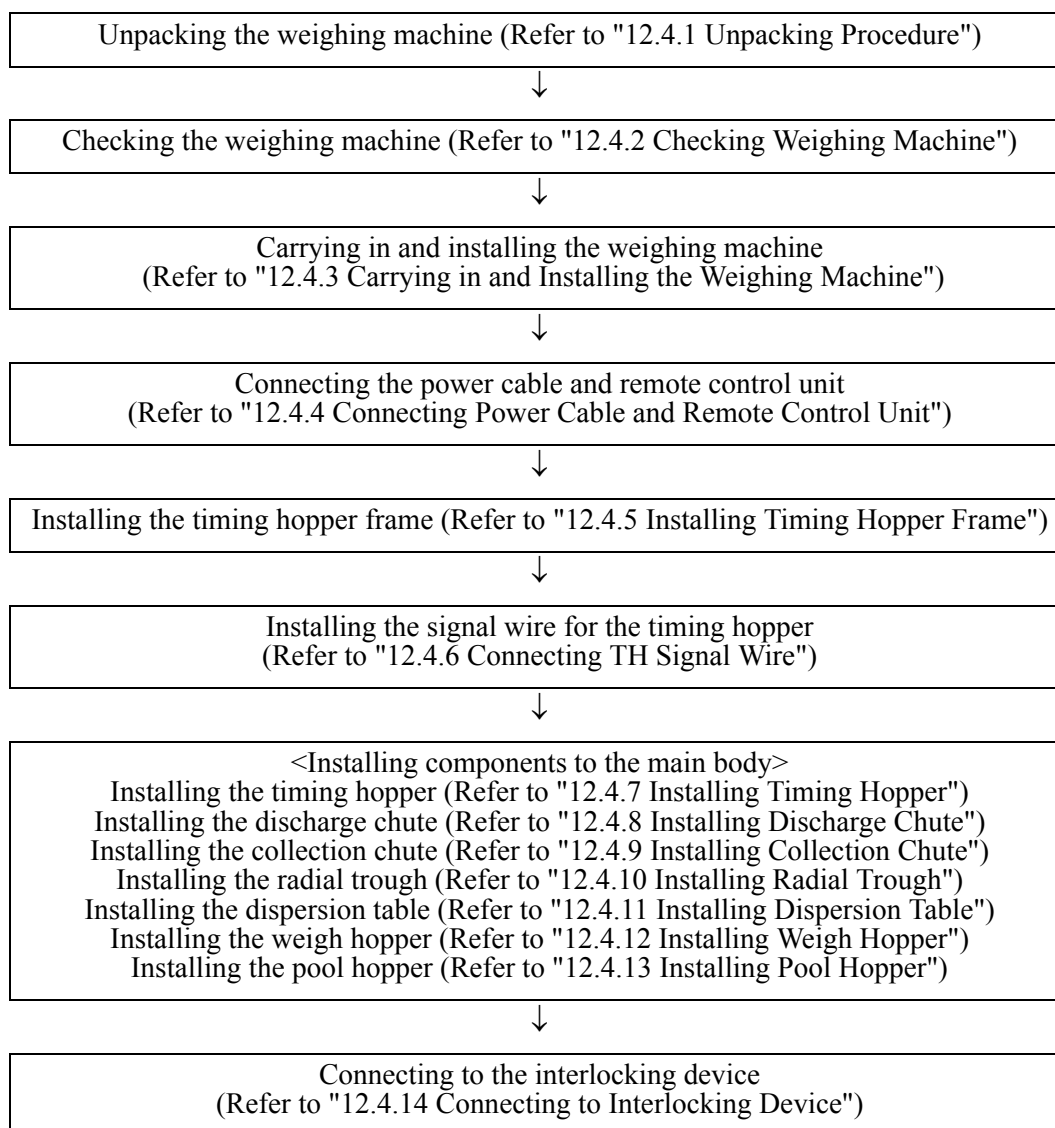
The overview for the weighing machine installation procedure is described below.
For details of each procedure, refer to the following reference section.

WARNING

- **The installation and test operation of the device can be performed only by the Ishida service department.**

CAUTION

- **When relocating the device, contact the distributor or Ishida Customer Support.**



NOTE

- After installing the weighing machine, perform check and adjustment of the installation condition (Refer to "12.5 Check and Adjustment After Installation").

12.4.1 Unpacking Procedure

**CAUTION**

- **When unpacking the weighing machine, wear protective gloves to avoid injury from the cardboard or crate.**

The weighing machine is packed in a cardboard box (For import only).
When unpacking, first remove the top lid of the package, then remove the side cardboard and crate.

12.4.2 Checking Weighing Machine

After unpacking, check the following items:

1. Damages to the weighing machine from shipping.
2. Parts shortage.

NOTE

- If there is any damage or shortage of the parts, contact the distributor or Ishida customer support.

12.4.3 Carrying in and Installing the Weighing Machine

**WARNING**

- **When lifting the weighing machine, follow the instructions in section 12.3.1 for safety. If they are not observed, injuries, fatal accidents or damage to the weighing machine may occur.**
- **When working on a scaffold, wear a safety belt in order to avoid a fall.**
- **When installing the weighing machine, ensure sufficient working space is provided around the weighing machine for cleaning, maintenance, inspections or daily production.**

Install the weighing machine using a crane or similar equipment to the scaffold (Refer to "12.3 Transportation and Lifting").

12.4.4 Connecting Power Cable and Remote Control Unit

After installing the weighing machine to the predetermined location, perform wire routing and installation position adjustment of the remote control unit in accordance with the following procedures.

WARNING

- **Connecting work between the weighing machine and the power supplies should be performed by a licensed electrician in accordance with the national and local electrical regulations.**
- **Install the remote control unit in the location that facilitates easy operation, free from any obstacle to the operation.**
- **Before connecting to the plant power terminal, use a tester to check if the power specifications of the weighing machine match with the plant power supply.**
When power of different specifications is accidentally supplied, serious damage to the weighing machine may occur.

1. Fix the main power switch in accordance with the following procedures.
 - a. Install the fixing metallic fitting (E) of the main power switch to the base metallic part of the remote control unit (A), using two screws (E1).
 - b. Fix the main power switch (B) to the fixing metallic fitting (E), using two screws (E2).
2. Adjust the remote control unit angle in accordance with the following procedures.
 - a. Fix the installation bracket (D) of the remote control unit to the specified position, using four screws (D1).
 - b. Adjust the remote control unit to an easily viewable angle, using the knob bolts (D3) and angle adjustment screws (D2), and fix it in the position.

CAUTION

- **Before connecting to the plant power terminal, make sure that the electricity is shut off, and then connect the power cable (H) to the plant power terminal. Before connecting to the plant power terminal, connect the remote control cable (F) and the power cable (G). Make sure that the power circuit breaker of the weighing machine is turned OFF.**

3. Guide the power cable (G) inside of the main body frame, and fix the flexible tube to the main body frame.
4. Connect the power cable (G) to terminals L11 and L31 of the terminal block XT601.
5. Guide the remote control cable (F) inside of the main body frame, and fix the flexible tube to the main body frame.
6. Connect the communication cable XJ02 of the remote control cable (F) to the relay connector of the CAL unit located in the main body frame.
7. Connect the RCU power cable XC21 of the remote control cable (F) to U01 of the PS-0 unit located in the main body frame. Connect the ground wire to the PS-0 unit grounding block.

8. Connect the customer power source side of the power cable (H) to the plant power terminal (outlet) that is closest to the weighing machine.

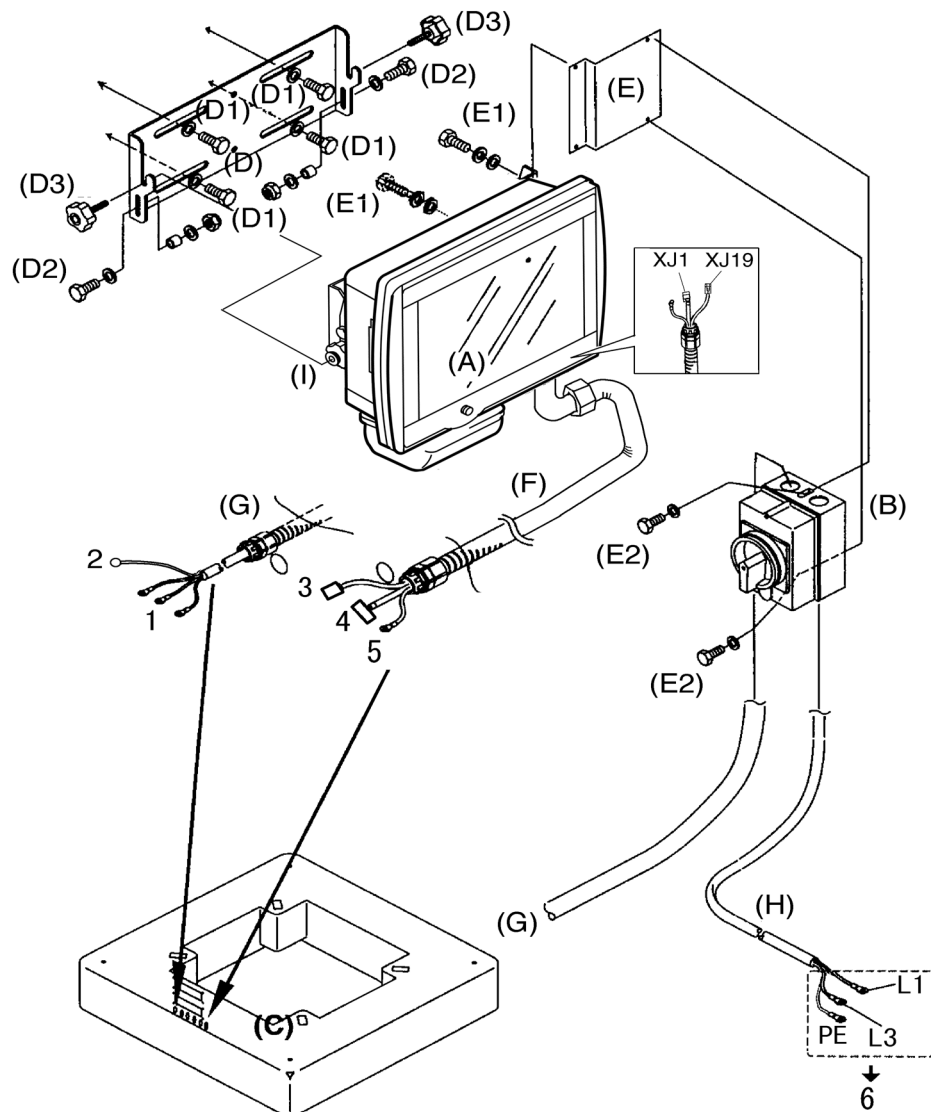


Fig. 12-2

Table 12-2

No.	Connecting Location
1	PS-0 unit (Terminal block XT601)
2	PS-0 unit (QF2, primary side)
3	PS-0 unit (U01-XC21 connector)
4	Ethernet relay connector
5	Ground terminal block
6	Customer power terminal

12.4.5 Installing Timing Hopper Frame

1. Install the timing hopper frame to the bottom of the main body, using four bolts.

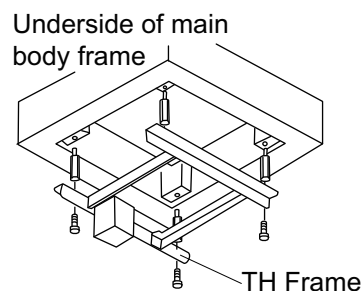


Fig. 12-3

12.4.6 Connecting TH Signal Wire

1. Fix the flexible tube for the timing hopper to the main body frame from inside of the main body frame.
2. Insert the TH signal wire to the relay connector of the DUC board.

NOTE

- Connect in accordance with the connector number of the TH signal wire, and the relay connector indication of the DUC board.

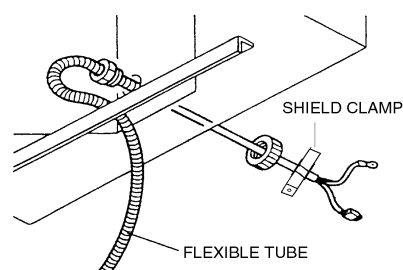


Fig. 12-4

3. Fix the shield coating portion of the TH signal wire to the main body with the shield clamp and bolts.

12.4.7 Installing Timing Hopper

1. Install the timing hopper installation shaft by hanging the shaft to the hook of the main body.
2. Check that the lever of the timing hopper fits appropriately.

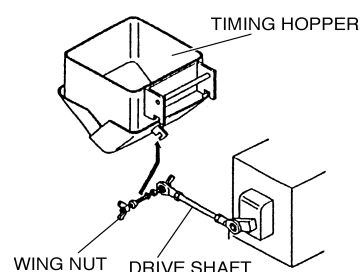


Fig. 12-5

12.4.8 Installing Discharge Chute

1. Insert the arm holes of the discharge chutes over the four supporting pins of the main body frame.

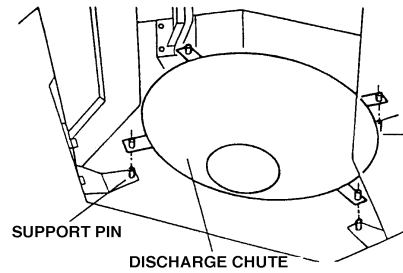


Fig. 12-6

12.4.9 Installing Collection Chute

1. Hang the hooks on both sides of the collection chute to the chute hanger of the main body frame support column.

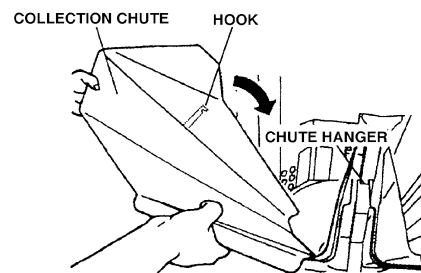


Fig. 12-7

12.4.10 Installing Radial Trough

1. Place the radial trough on the trough base.
2. Tighten the clamp lever to fix the radial trough.

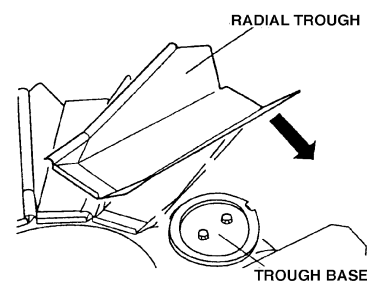


Fig. 12-8

12.4.11 Installing Dispersion Table

1. Screw the dispersion table into the feeder.

NOTE

- For some types of the weighing machine, the dispersion tables are fixed with bolts.

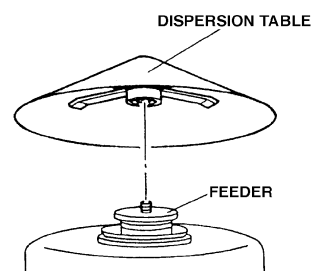


Fig. 12-9

12.4.12 Installing Weigh Hopper

CAUTION

- When installing the weigh hopper, do not give a shock to the weigh hopper hanger of the weigh drive unit. Doing so may cause malfunction of the weigh sensor.

NOTE

- The weigh hoppers do not have matching numbers. The weigh hoppers can be installed to any of the heads.

1. To install the weigh hopper, apply the hooks of the installation portions to the installation pins of the main unit.
2. Make sure that the open/close roller of the weigh hopper fits securely into the switching lever of the main body.

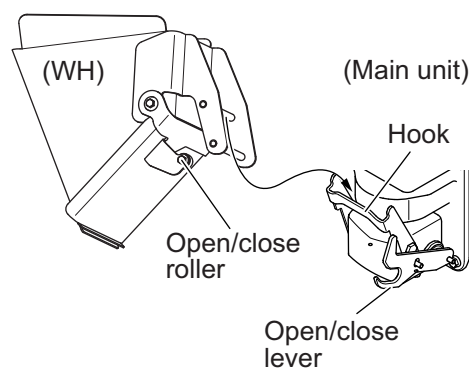


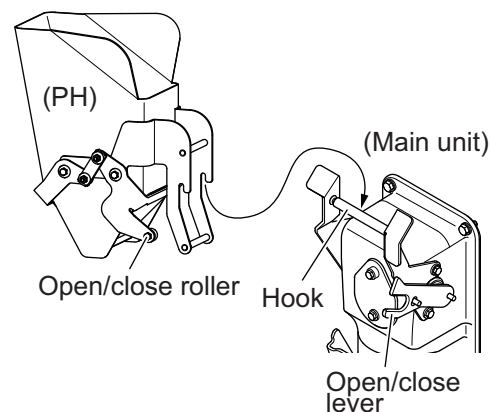
Fig. 12-10

12.4.13 Installing Pool Hopper

NOTE

- The pool hoppers do not have matching numbers. The pool hoppers can be installed to any of the heads.

1. To install the pool hopper, apply the hooks of the installation portions to the installation pins of the main unit.
2. Make sure that the open/close roller of the pool hopper fits securely into the switching lever of the main body.

**Fig. 12-11**

12.4.14 Connecting to Interlocking Device

Connect the interlock signal wires of the relay unit to the interlocking devices such as packer and feeder.

! DANGER

- All wiring inside the main body, control panel or operation panel can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

! WARNING

- When other devices need to be connected before and after this weighing machine, shut off the power for all of the devices. Failure to do so may cause an electrical shock.

! CAUTION

- In order to prevent malfunction of the weighing machine, route the interlock signal wire away from the power cables. When using relays and contactors as a load, prevent induction noise by using a surge killer or other methods.

1. Specifications of the interlocking device signal

For the interlocking device signals, follow the specifications below.

Table 12-3

Interlock Target	Signal	Input/Output Characteristics of the Weighing Machine
1.Packer	Interlock signal. The interlock signal from the packer to the weigher should output a no-voltage contact signal.	Input characteristics. Circuit: opto-isolator. Voltage: 24 VDC. Current: approx. 15 mA. Signal width: min. 50 msec.
	Discharge completion signal. The discharge completion signal from the weigher to the packer outputs a no-voltage contact signal.	Output characteristics. Circuit: relay contact. Maximum Rating: 5 A at 250 VAC Signal width: set arbitrary with the remote control. (Default value is 100 msec)
2.Infeeder	Infeed control signal. The feed signal from the weigher outputs a no-voltage contact signal.	Output characteristics. Circuit: relay contact. Maximum Rating: 5 A at 250 VAC
3.Other	Error signal. The error signal from the weigher outputs a no-voltage contact signal.	Output characteristics. Circuit: relay contact. Maximum Rating: 5 A at 250 VAC

2. Signal name and connecting location of relay unit external connection terminal

The signal names and connecting locations of the external connection terminals of the relay unit are as follows.

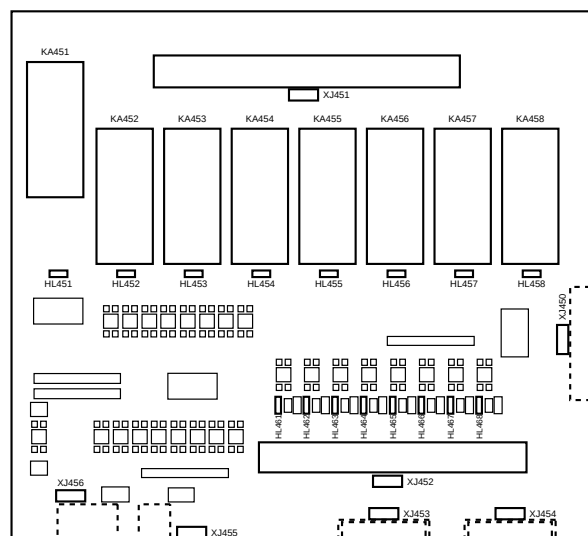


Fig. 12-12

Table 12-4

Terminal No.	Terminal Block No.	Signal Name	Connecting Location
28-29	XJ451/1-2	Discharge completion signal 1	Packer
78-79	XJ451/3-4	Discharge completion signal 2	Packer
82-83	XJ451/5-6	Error signal 1	—
90-91	XJ451/7-8	Error signal 2	—
30-31	XJ451/9-10	Infeed control signal 1	Infeeder
88-89	XJ451/11-12	Infeed control signal 2	Infeeder
84-85	XJ451/13-14	Control 1	* Optional
86-87	XJ451/15-16	Control 2	* Optional
20-21	XJ452/1-2	Interlock signal 1	Packer
21-72	XJ452/2-3	Interlock signal 2	Packer
22-21	XJ452/4-5	Input signal 5	* Optional
21-73	XJ452/5-6	Input signal 6	* Optional
25-26	XJ452/7-8	Input signal 1	* Optional
26-102	XJ452/8-9	Input signal 2	* Optional
104-105	XJ452/10-11	Input signal 3	* Optional
105-106	XJ452/11-12	Input signal 4	* Optional
53-54	XJ452/13-14	DC+24 V	—

12.5 Check and Adjustment After Installation

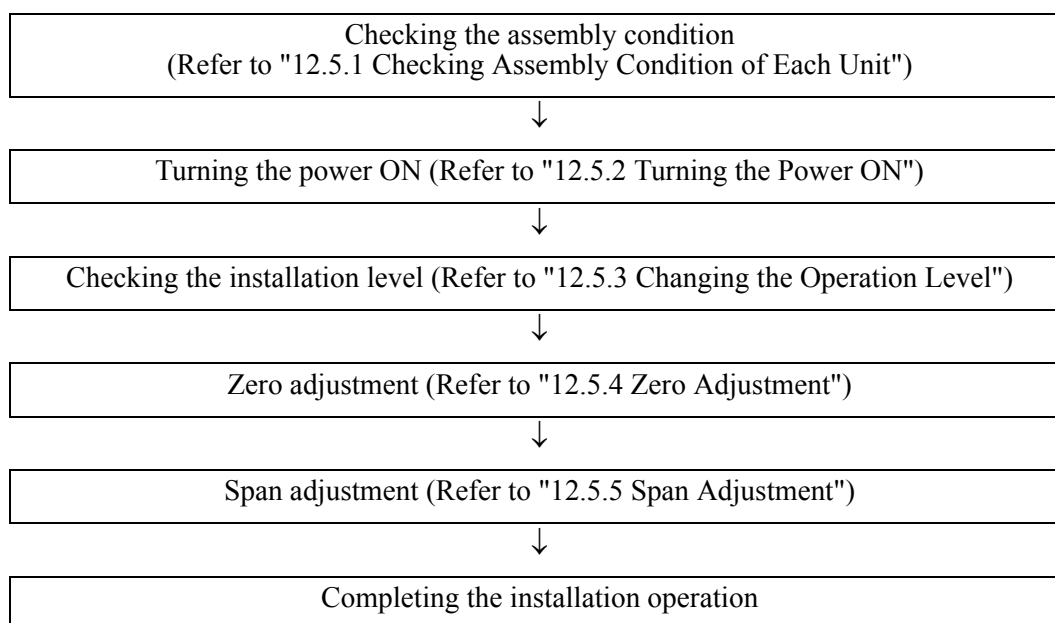
After installing the unit and components, and routing the wiring, check the installation condition.

WARNING

- The installation and test operation of the device can be performed only by the Ishida service department.
- When entering the weighing machine to check the installation condition, the operator must turn OFF and lock the main power switch, and keep the key in his possession during the work.

CAUTION

- When relocating the device, contact the distributor or Ishida Customer Support.



12.5.1 Checking Assembly Condition of Each Unit

Check that each unit such as the dispersion table, radial trough, discharge chute and hoppers are securely installed in accordance with the following procedures:

1. Checking secure installation of the pool hopper
Make sure that the open/close roller of the pool hopper fits securely into the switching lever of the main body.

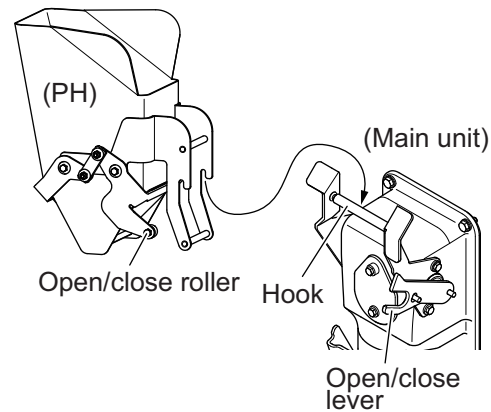


Fig. 12-13

2. Checking secure installation of the weigh hopper
Make sure that the open/close roller of the weigh hopper fits securely into the switching lever of the main body.

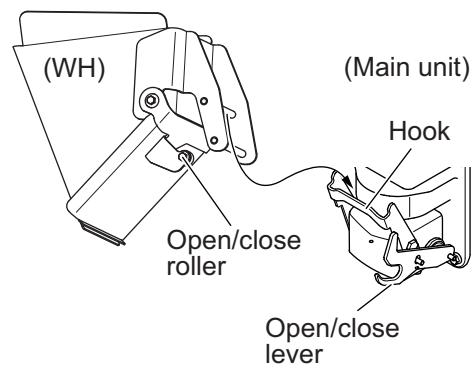


Fig. 12-14

3. Checking secure installation of the dispersion table
 - Make sure that the dispersion table is fixed securely.
 - Make sure that there is no interference between the dispersion table and radial trough.

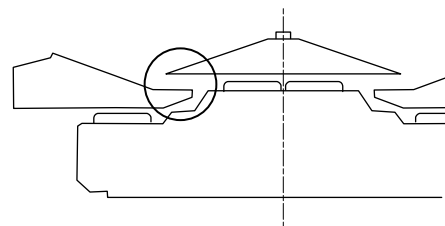


Fig. 12-15

4. Checking for the interference between radial troughs
Make sure that there is no interference between radial troughs.
If there is interference, reinstall the radial troughs.

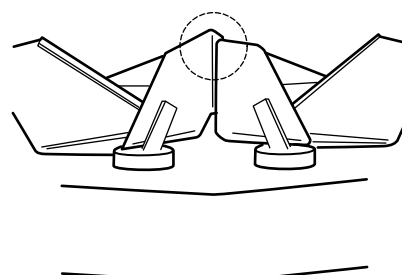


Fig. 12-16

12.5.2 Turning the Power ON

CAUTION

- Before connecting to the plant power terminal, use a tester to check if the power specifications of the weighing machine match with the plant power supply.
When power of different specifications is accidentally supplied, serious damage to the weighing machine may occur.

Turn the power ON in accordance with the following procedures.

Connect the power outlet, and turn the main power switch next to the remote control unit ON. The Main Menu of the Operator level is displayed automatically.

1. Check the wiring of terminal block PS-0.
2. Check that the breaker is ON.

NOTE

- Keep the breaker inside the main body ON all of the time.

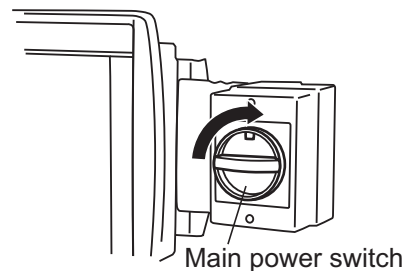


Fig. 12-17

3. Turn ON the main power switch of the factory side, and turn ON the main power switch next to the remote control unit.

► The Main Menu appears.

4. Press the [Power] key.

► The power will be supplied to the hopper drive circuit and feeder drive circuit.



Fig. 12-18

12.5.3 Changing the Operation Level

Change to the Installation level in accordance with the following procedures.

1. Press the [Operation Level Selection] key.

- ▶ The Operation Level Selection menu appears.



Fig. 12-19

2. Select "Installation".

- ▶ The Password Input Keyboard screen appears.

3. Enter the password.

- ▶ The Main Menu for the Installation level appears.



Fig. 12-20

NOTE

- The factory default password set for the Installation level is "2".
- The entered password is displayed as "*".

12.5.4 Zero Adjustment

Check the Installation level in accordance with the following procedures.

1. Press the [Machine Set] pop-up key.
 ► The Machine Set pop-up menu appears.



Fig. 12-21

2. Select "Manual Adjustment".
 ► The Manual Adjustment screen appears.
3. If the [Weighing Adjst] tab is not selected, press the [Weighing Adjst] tab.
 ► The weight of each head is displayed.



Fig. 12-22

4. Press the [ALL Head SLCT/CLR] key.
 ► All heads are selected.



Fig. 12-23

5. Press the [Zero Adjst] key.

- The hopper opens/closes, and along with the message "Please wait a moment," the zero adjustment starts.



Fig. 12-24

6. After the zero adjustment is complete, check that the weight display of each weigh head is within $0.0\text{ g} \pm 0.1\text{ g}$.

NOTE

- If the weight display of each head exceeds 0.1 g or falls below -0.1 g , repeat the procedure from step 3.
- The operation details of the zero adjustment procedures (Refer to "6.5 Zero Adjustment screen")
- Then, perform the span adjustment (See the following.).

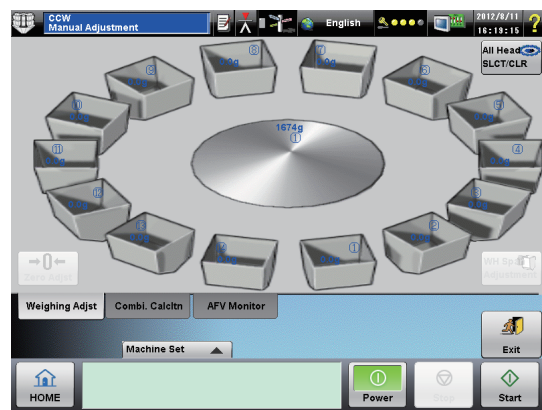


Fig. 12-25

12.5.5 Span Adjustment

For the span adjustment, follow the procedures below.

1. Place the span adjustment weight on all of the weigh hoppers.

NOTE

- Normally the span adjustment weight is 200 g. However, it may vary depending on the specifications.

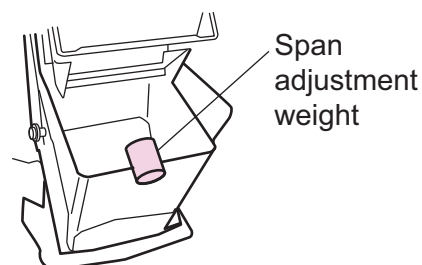


Fig. 12-26

2. Press the [ALL Head SLCT/CLR] key.

► All heads are selected.



Fig. 12-27

3. Press the [WH Span Adjustment] key.

► The span adjustment starts.



Fig. 12-28

4. Make sure that the weight display of each head is within $200.0 \text{ g} \pm 0.1 \text{ g}$.

NOTE

- Make sure that the weight display of each head does not exceed $200.0 \text{ g} \pm 0.1 \text{ g}$. If it exceeds 200.1 g , or falls below 199.9 g , repeat from step 2.



Fig. 12-29

5. Remove the span adjustment weight from the weigh hopper.

- Make sure that the weight display of each head is within $0.0 \text{ g} \pm 0.1 \text{ g}$.

NOTE

- If the weight display of each head exceeds 0.1 g , or falls below -0.1 g , repeat from step 3 of "12.5.4 Zero Adjustment".

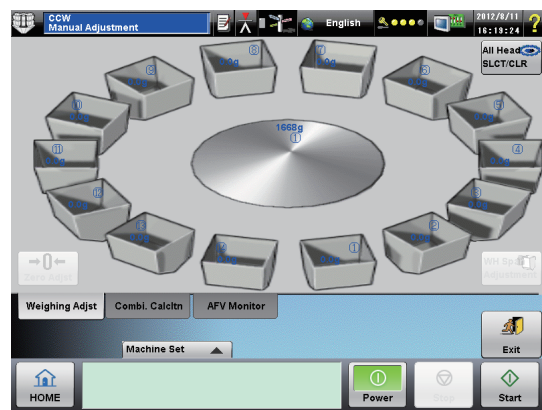


Fig. 12-30

6. Press the [Exit] key.

- The display returns to the Main Menu.

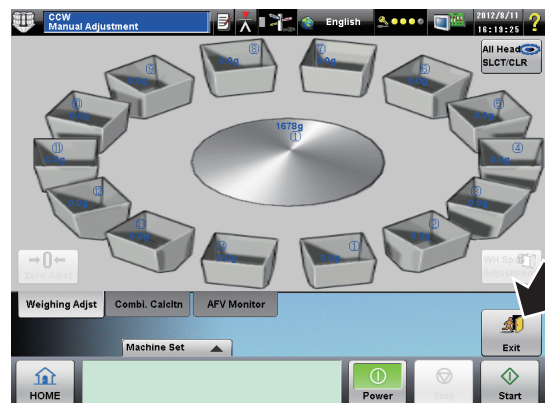


Fig. 12-31

12.5.6 Check Items at Installation

DANGER

- All inspection and servicing of the main body, control panel or operation panel can be performed only by Ishida service department. Contact the distributor or Ishida Customer Support.

Check the following items during installation.

Table 12-5

Item	Contents	Procedures	Criteria
1.Check model	1.Compare machine with type (model) written in the instructions.	Check with model block diagram and standard specifications.	• They should match (Check visually).
2.Check power	1.Voltage and frequency.	Verify with mechanical, electrical, and manufacturing instructions and requests.	• They should match (Check visually).
3.Check vicinity of main body	1. Check breaker.	Press button.	• Check operation panel display visually.
	2.Check wiring of terminal block PS-0 and PS-2.	Pull the code using hands (Do not pull strongly).	• Code should not be pulled out.

Table 12-5

Item	Contents	Procedures	Criteria
4. Check operations (Continue to next page)	1. Check operation panel display.	Check for errors on display.	No display fluctuation or missing dots detected.
	2. Check USB.	Insert the USB to the remote control BOX, and write/read out presettings.	Check with remote control display.
	3. Check printer.	Set to preset, and press [Output] key to print out.	Display and printed contents should match.
	4. Check zero adjustment.	Perform zero adjustment, and check zero point of each head.	Weigher is in stable condition, and zero point is within ± 0.1 g (± 1 graduation).
	5. Check span adjustment.	After zero adjustment, place span adjustment weight on each head, and perform span adjustment. <u>Cell Span adjustment weight</u> Standard capacity spec. 6 L 200.0 g	Weigher is in stable condition, and span value is within ± 1 graduation. <u>Span adjustment value</u> 200.0 ± 0.1 g
	6. Check radial feeder and dispersion feeder operations.	Put weigher into operation.	- Vibrate (Check with hands). - At feeder value of 99, vibration display is within 3.0 ± 0.25 mm at strong vibration.

Table 12-5

Item	Contents	Procedures	Criteria
4. Check operations (Continued)	7. Check smooth flow between pool hopper and weigh hopper.	Check parallelism between pool hopper and weigh hopper. Open the gate of the pool hopper and check for a smooth flow.	• Difference between left and right should be within 1.5 mm. • Pool hopper gate does not make contact with weigh hopper.
	8. Check operations and stop positions of pool hopper lever and weigh hopper lever.	Put weigher into operation, place weights on each weigh hopper sequentially to activate pool hopper and weigh hopper.	• Check for the smooth operation of each lever. • Check if it stops at the home position.
	9. Check gate noise and opening condition of pool hopper and weigh hopper.	Put weigher into operation, place weights on each weigh hopper sequentially to activate pool hopper and weigh hopper.	• Check that noise is at a minimum when the gate is closed. • There should be no metallic sound.
	10. Check actual weighed value and display value.	Set to print the weighing result every time.	• Actual weighed value ± 5 graduations or more should not be printed more than twice in a row.
5. Check external signals	1. Check interlock signal.	Short-circuit interlock signal wire.	• Relay board LED should light up.
	2. Check discharge completion signal.	Place weights on weigh hopper according to set value, and perform discharging operations.	• Check with relay board if LED lights up or by tester.
	3. Check infeed control signal.	Press dispersion table by hand.	• Check with relay board if LED lights up or by tester.
6. Check errors	1. Check for incorrect zero point.	Set auto zero interval, and perform discharging operations.	• Verify zero error and head number with remote control display. • Check with relay board if LED lights up.
	2. Check error with pool hopper.	Remove pool hopper, and press switching lever of pool hopper using hands during the discharge operation.	• Verify pool hopper error and head number with remote control display.
	3. Check error with weigh hopper.	Remove weigh hopper, and press switching lever of weigh hopper using hands during the discharge operation.	• Verify weigh hopper error and head number with the remote control display.

Table 12-5

Item	Contents	Procedures	Criteria
7. Check documents	1.Check mechanical manufacturing instructions.	—	• Verify visually.
	2.Check electrical manufacturing instructions.	—	• Verify visually.
	3.Check manufacturing requests and order specifications.	Check main model of branch number, and specifications.	• Verify visually.
	4.Check specifications.	—	• Verify visually.
	5.Check layout diagram.	Installation position and dimensions of remote control BOX. (Height, width, discharge height, etc.)	• Check visually and by actual measurement.
	6.Check attachments and spare parts.	Check tool box, weights, weigher, and attachments.	—